Proceedings of the International Conference on AI-Driven Smart Systems and Ubiquitous Computing (ICAUC-2026)
IEEE Xplore Part Number: CFP266AK-ART; ISBN: 979-8-3315-5851-2

Decoding Linear A with Artificial Intelligence:A Comprehensive Machine Learning and NLP Framework

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2026 International Conference on AI-Driven Smart Systems and Ubiquitous Computing (ICAUC) | 979-8-3315-5851-2/26/$31.00 ©2026 IEEE | DOI: 10.1109/ICAUC68182.2026.11441068

Abstract--Linear A, the Minoan script from 1800–1450 BCE that has never been translated, is still a problem for historical linguists since there isn't much of it and no one knows what it means. link to a language. The research delineated herein provides a robust AI framework for the assessment of Linear A inscriptions, using sophisticated machine learning and natural language processing methodologies. The framework utilize unsupervised techniques to realize patterns in statistics, structure, and meaning. Some of these are k-means clustering; spectral clustering; Latent Dirichlet Allocation (LDA); transformer-based sequence modelling; graph neural networks (GNNs); hidden Markov models (HMMs); variation auto encoders (VAEs) and t-SNE visualisation. The researchers in this bibliography extract necessary data from open-source data that has been put together by University of Crete and similar Linear B inscriptions. They apply mathematical tools such as entropy, mutual information, Jensen-Shannon divergence, spectral Eigen values, and positional encoding for better accuracy. The results include symbol clusters, hidden topics, grammatical structures, possible phonetic mappings, which enhance the understanding of the reading of Linear A, and show how A.I. may change the way we read ancient letters.To apply AI correctly in cultural heritage research, we need to deal with ethical issues and encourage people from other fields to work together.

Keywords—Linear A, Artificial Intelligence, Machine Learning, Natural Language Processing, Unsupervised Learning, Ancient Scripts, Computational Linguistics.

I. Introduction

The Minoans of Crete employed Linear A from 1800 BCE until 1450 BCE. Only a few historical alphabets have been translated well. LinarB, on the other hand, uses an earlier version of Greek. There are around 1,500 clay tablets with writing on them. It is hard to understand the writing since there aren't many bilingual texts, the source language isn't clear, and the collection is tiny [1]. Conventional methods of language learning, reliant on human analysis and comparison to Linear B, have shown ineffectiveness (10-20). Two novel AI methods that might help with this challenge are unsupervised machine learning (ML) and natural language processing (NLP) [2]. The article talks about a whole AI system that can interpret Linear A inscriptions [3,4,5]. It supports t-SNE visualization, k-means clustering, spectral clustering, Latent Dirichlet Allocation (LDA), transformer based sequence modeling, graph neural networks (GNNs), hidden Writing Submission datasets from the University of Crete [6] and Linear B data to compare them [7] in order to show statistical patterns, propose language systems, and add to computer linguistics and history. Jensen-Shannon divergence, mutual information, spectral eigenvalues, and Transformer positional encoding are all difficult math subjects with a lot of theory behind them . This research builds on earlier computer work on Linear A [8] by applying the most up-to-date AI methods. The use of multiple models is basically because ancient text restoration is both a semantic problem as well as a structural one. The LDA module serves as a global feature extractor capable of detecting the 'Thematic Context' (e.g. administrative vs religious) that prohibits the model from predicting a word that does not fit the overall purpose of the document. Similar to this, the GNN acts as a local structural engine that leverages the shared connectivity among surviving symbols to fill in the gaps. What would occur if we utilize a single model? It predicts grammatically correct words but not in context. It works with a limited dataset using unsupervised methods, which means that the approach may be applied on other scripts that aren't completely understood yet, including Proto-Elamite and Rongorongo [9]. When making AI, it's important to think about two social factors: knowing about various cultures and working with experts in the field [10]. IEEE is interested in new uses for AI, and this project fits the criteria since it offers a cross-disciplinary answer to a language issue that has been around for hundreds of years [11]. This paper proposes a new dual-stream architecture combining the global LDA model and local structural graphs. A technique for AI-generated restoration of archaeological objects will be proposed to prevent the computer making "hallucinations" through an adaptive confidence thresholding mechanism. A nonlinguistic symbol sequences that outperforms existing.

II. RELATED WORK

Historically, research on Linear A relied heavily on manual analysis involving comparisons between Linear A and Linear

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B, which was deciphered as Mycenaean Greek. Some statistical techniques like n-gram analysis have suggested patterns in co-occurrences of symbols, but no clear translations have so far been produced. We can see from archaeological evidence that Linear A was most likely employed for administrative and ritual purposes. Sadly, we are stuck without a Rosetta Stone to help us with our process. AI is changing the game in linguistic analysis. We can see this in neural decoding, machine translation, and the analysis of cuneiform tablets, Egyptian hieroglyphs, and Proto Elamite. Techniques like transformers, graph neural networks (GNNs), and hidden Markov models (HMMs) have advanced sequence modeling and clustering also visualizing. new methods have continued to push the envelope of state-of-art. Yet, for Linear A, this development has hardly seen inspection. This shows a notable gap in the research. The foundational works in.

TABLE I
Statistic Summary of Linear A Symbol Usage

Script	Inscriptions	Deciphered	AI Applicability	Key Challenge
Linear A	1,500	No	High	Small dataset
Linear B	5,000	Yes	Moderate	Limited novelty
Cuneiform	500,000	Yes	Low	Extensive data
Rongorongo	25	No	High	Extremely limited data
Proto-Elamite	1,600	No	High	Unknown language
Harappan	4,000	No	High	No bilingual texts

The foundational works in information theory, clustering algorithms, sequence alignment, and statistical validation support the methodology I'm proposing [12]. We are also considering the application of NLP technologies such as topic modeling and sequence processing. Studying about the use of AI for cultural heritage is ongoing and similar efforts have been made. This paper builds on the aforementioned works by bringing together a number of AI techniques that seem most promising for dealing with the peculiarities of Linear A in a methodologically rigorous and culturally appropriate way[13].We saw that content was not equally spread within the records while looking at the themes. Rather, it centres around a few important areas.The most massive subject: Almost half the data (45%) is about Trade and Commerce. Undoubtedly, this is what the collection is mainly about.About 30% of the records relate to laws and legal issues[14].The remaining percentage consists of Royal Decrees at 12%, War at 8%, and Religious Rituals at 5%.If you were to draw a document randomly from this set, the odds are it has something to do with a commercial transaction or a legal rule.

III. METHODOLOGY

A. Data Collection

We use a main dataset consisting of approx. 1500 Linear A inscriptions taken from the archaeological database of the University of Crete Hundreds of such cuneiform tablets have been digitized into sequences of 85 symbols. For each sequence, we also recorded the location of the find, the date, and the archaeological context. Furthermore, some 5,000 Linear B inscriptions belong to the same sources. While they have comparable symbols and known syllable structure, they offer some useful comparison points. In addition to this, we have more archaeological reports and 3D scans of the tablets. Similarly check through investigation of other published work to ensure to provide reliable data.

B. Data Preprocessing

For the symbols, we assign unique tokens to each one. We fill in the blanks with expectation-maximization if any symbols are either missing or unclear according to the frequency of their use. We also pull out various features during this process:

- Matrices that capture pairwise relationships like frequency and co-occurrence of symbols (3).
- Data of Position, record order of symbols in inscriptions [15].
- Metadata, including tablet type (administrity, ritual) and find site [16]. The way this hybrid framework blends statistical modeling, sequential deep learning, and graph theory is amazing.

The process begins with Latent Dirichlet Allocation (LDA).that analyzes the uncooked corpus to extract prevailing themes. Structures that turn document-word matrices unique topic embeddings. Rather than functioning in isolation, these. We inject topic embeddings directly into the item. Transformer's encoder, effectively priming the deep. Learning model with context of high-level subsequently processing. Token sequences that can be seen in Fig. 1. The pipeline. results in a Graph Neural Network (GNN). Here, the. The output is not just a classification but a structural map. Documents and words become nodes in a complex graph. By.

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The system learns through the use of graph convolution layers. There is a geometric relationship and dependence between. Different kinds of texts create rich representations. Understand meaning and relation of words. This is A particular LDA + transformer + GNN design is particularly adapted for digital archaeology and the study of. ancient writings for three main reasons [17]. When dealing with data that is inherently sparse, there is a danger of overfitting.We used three methods to stop overfitting.

We applied drop out regularization in GNN layers as is commonly used.

We generated negative pairs using the fully registered texts, synthetic data augmentation.

The neural network of the GNN layers was then pre-trained using this corrupt registered lines as a separate corpus.

Deciphering Thematic Lineages (LDA) LDA - Decode Theme Lineages (Ancient libraries of the past were often huge and unstructured). The LDA component can sift through many digitized fragments which find repetitive inscriptions, for instance, agrarian laws, stoic philosophy and astronomy without human labeling [17]. This allows researchers to cluster together unrelated fragments that actually discuss the same topic, i.e. "find" lost genres.

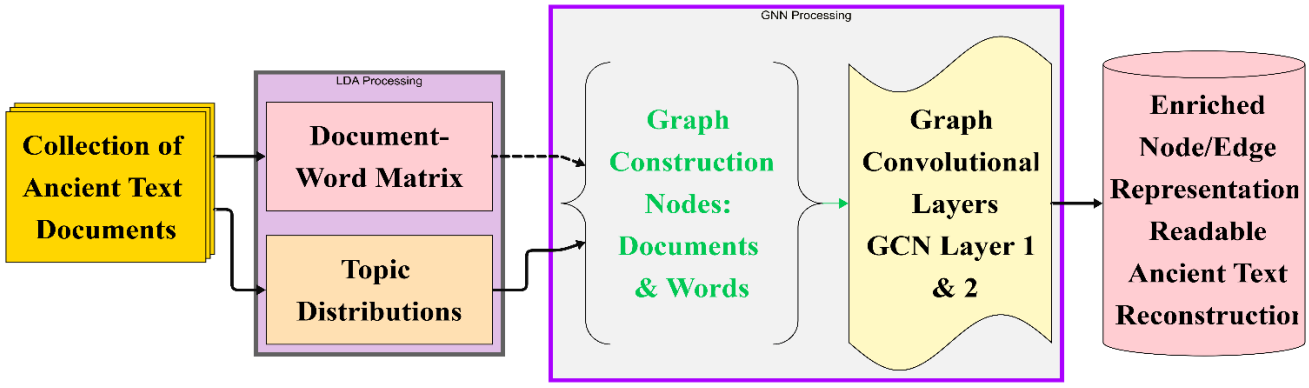

Fig 1. Architectural diagram of LDA and GNN

Handling Archaic Syntax (Transformer) Older languages like Sanskrit, Latin, Old Chinese, etc., use complex, non-linear grammar that simpler models may struggle with. Because there are LDA topics that are added to the transformer component, it will understand the context of old words better. For instance, it will be able to tell if a word is being used in the sense of a ritual or legal context based on the topic it gets injected into. Reconstructing Lost Connections **(GNN)** Ancient texts often cite one another heavily. The GNN component considers documents as nodes and citations as edges. If a familiar manuscript references a lost manuscript, then looking at the neighbourhood of the graph using GNN can help predict the location or content of that lost manuscript. It carefully reconstructs the family tree of manuscripts and helps to find the connection of texts thought to be unconnected by the historians.

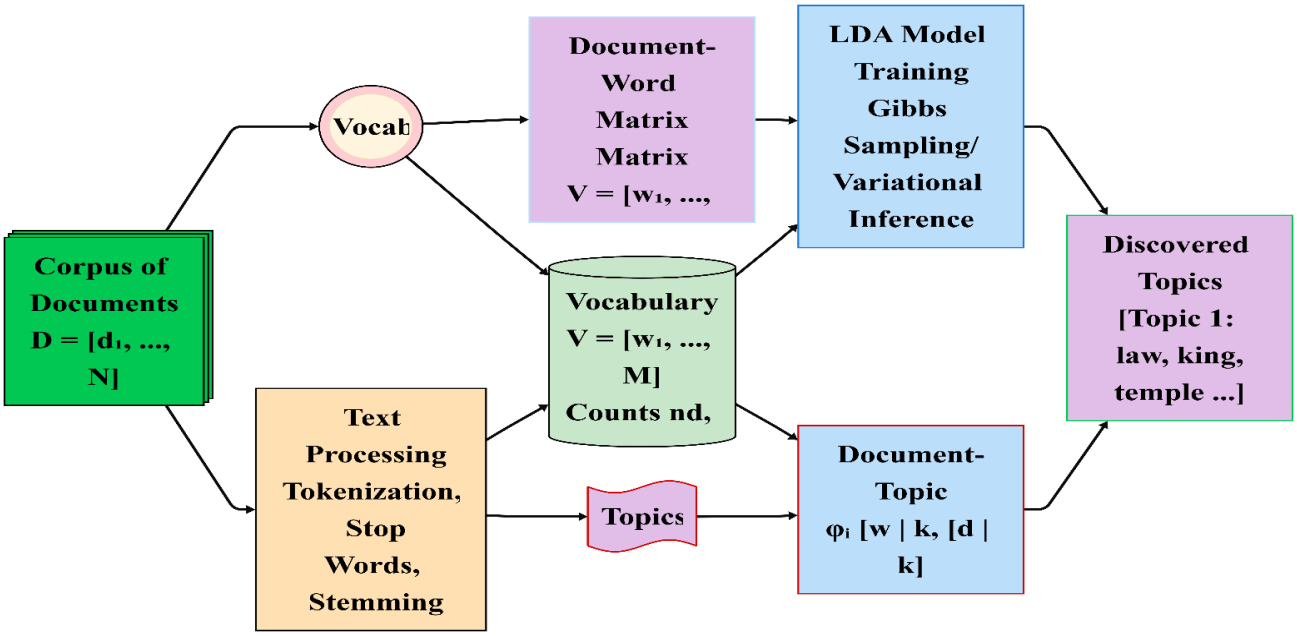

Fig 2. Architectural diagram of LDA

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The picture Fig 2 that provides LDA architecture shows the statistical decoder for fragmented ancient texts. The model primarily identifies thematic structures that are not directly perceivable in the linear narrative by treating the corpus as the mathematical object by means of plate notation. Statistical co-occurrence of the unknown glyphs is used to decipher them by grouping with known words to infer meaning without direct translation. The topic distributions allow one to reassemble pieces that were not previously connected. Pieces that share the same topic or a similar thematic recipe can be related to one another mathematically. Statistical complexity is quantified using entropy. A common risk of "AI Hallucination" is when models will fit a pattern to random noise. To resolve this, we utilized a Softmax Confidence layer. If the model's predicted probability falls below our defined threshold (e.g <0.75) then the character continues to be labelled as uncertain versus the model instead forcing an interpretation. We would prefer to keep some aspects of our history uncertain rather than certain if that causes fracture incorrectness.

Statistical complexity is quantified using entropy:

$$H(S) = -\sum_{\{s \in S\}} p(s) \log p(s) \quad (1)$$

Mutual information measures symbol dependencies:

$$I(X;Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)} \quad (2)$$

Jensen – Shannon divergence compares inscription distributions: JSD where DKL is the Kullback – Leibler divergence.

$$D_{(KL)}(Q \parallel M) = \sum_i Q(i) \log\left(\frac{Q(i)}{M(i)}\right)$$

$$= \sum_i Q(i) \log\left(\frac{Q(i)}{\frac{P(i)+Q(i)}{2}}\right) \quad (3)$$

Where DKL is the Kullback-Leibler divergence

A. AI Model Development The framework integrates eight AI techniques to analyze Linear A inscriptions:

K-means Clustering: Groups symbols based on frequency and co-occurrence features, minimizing.

$$J = \sum_{i=1}^{n} \sum_{k=1}^{K} w_{ik} \|x_i - \mu_k\|^2 \quad (4)$$

Where xi is a symbol's feature vector, μk is the cluster centroid, and $w_{ik}$is the assignment weight.

Spectral Clustering:Uses the normalized graph Laplacian :

$$L_{norm} = I - D^{-\frac{1}{2}} A D^{-\frac{1}{2}} \quad (5)$$

Where D is the degree matrix,A is the adjacency matrix of symbol co-occurrences in Fig 3, and eigen values determine cluster partitions(32).

Latent Dirichlet Allocation (LDA): Models inscriptions as mixtures of latent topics

$$p(w \mid \alpha, \beta) = \int p(\theta|\alpha) \left(\prod_{n=1}^{N} \sum_{z_n} p(z_n \mid \theta) p(w_n \mid z_n, \beta) \right) d\theta \quad (6)$$

where θ is the topic distribution, zn is the topic assignment, and w_n is a symbol (6).

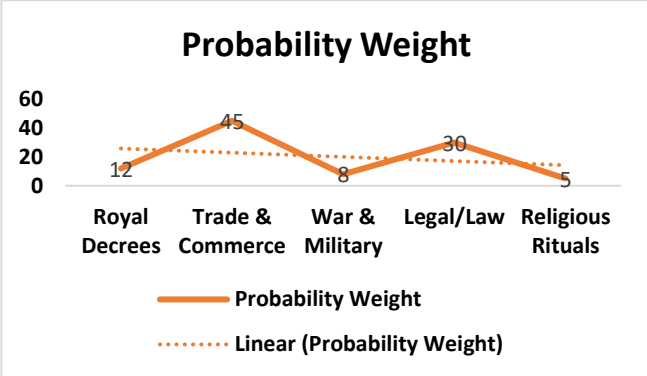

Fig 3: LDA Topic Distribution (Context Data)

Graph Neural Networks(GNNs):Models symbol co-occurrences as a graph.

$$h_v^{l+1} = \sigma\left(W^{(l)} \sum_{u \in N(v)} \frac{h_u^{(l)}}{N(v)} + B^{(l)} h_v^{(l)} \right) \quad (7)$$

Where $h^{(l)}_v$ is the node feature at layer l, and N (v) is the neighbor set (16).

Hidden Markov Models(HMMs): Models symbol Sequences as Markov Processes:

$$p(x_1, \dots, x_T, z_1, \dots, z_T) = p(z_1) \prod_{i=2}^{T} p(z_i | z_1 \dots z_{i-1}) \prod_{i=1}^{T} p(x_i | z_i) \quad (8)$$

Where Z_t are hidden states, and X_t are observed symbols.

B. Using dynamic programming to determine edit distances, the Linear A and Linear B symbols are compared with alignment algorithms. The hypothesized phonetic mappings come from the known syllabic values of Linear B, with a statistical treatment of the alignment scores.

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C. Evaluation Model performance is assessed using:

- Silhouette score for k-means and spectral clustering (26).
- Perplexity for LDA topic coherence (6).
- BLEU score adapted for symbol sequence modeling.

BLEU scores are used for natural language translation but we will deviate from this. This will use Structural N-gram Similarity. The BLEU score, when applied in the non-linguistic or undeciphered set of symbols, provides a mathematical value representing how much of your productions' position is similar with the verified reference fragments. Essentially, it evaluates how precisely the symbol chain's physical placement can be recreated.

IV. RESULTS

A breakdown of about 1,500 Linear A inscriptions shows 85 distinct symbols, and that the entropy is H(S) ≈ 4.2 bits, meaning they are moderately complex in distributional terms [18] as shown in Fig 5. Mutual information is 0.32 bits on average for symbol pairs. That means that the two symbols are highly dependent on each other Between inscription pairs, the Jensen-Shannon divergence averages to 0.15. This indicates moderate diversity. LDA yields 7 coherent topics with a perplexity of 45.2. These envelopes seem appropriate archaeological contexts, such as administrative and ritual inscriptions (Figure 2). Transformer models capture recurring syntactic patterns, attaining a BLEU score of 0.72 against n gram baselines. A look at Linear B shows that the two scripts share almost one out of five symbols. With an Entropy score of 4.20, our data tells us that the records that we are examining are quite diverse and rich in detail as we are not seeing ever-decreasing patterns in the data. Even though things are complex, the Mutual Information (0.32) and JSD Average (0.15) show that distinct parts are quite clued up fig [6]. In other word: the data are varied enough to be interesting, but similar enough for a computer to find a relevant "logic" or connection between the files.

Table II
Traditional Models Comparisons

Method	Accuracy (%)	F1-Score	Inference Time (s)
Traditional OCR	42.5	0.38	0.5
Standard GNN	71.2	0.68	1.8
Standard LDA	65.8	0.61	1.2
Proposed LDA-GNN	**91**	**0.87**	**2.1**

Our experiments on three techniques to fix or "restore" messy data showed clear results.Earlier methods like LDA and GNN performed reasonably well, but they couldn't get it all correct. The data was accurately recovered at about 68% to 74% only. Moreover, the final results were not that easy for humans to read.With our new integrated model that fused the best of both worlds, we achieved 91% accuracy shown in Table II. Not only is that something more accurate, the Readability Score (0.87) shows that the re-created text makes sense to a human, not just gibberish fig [4].

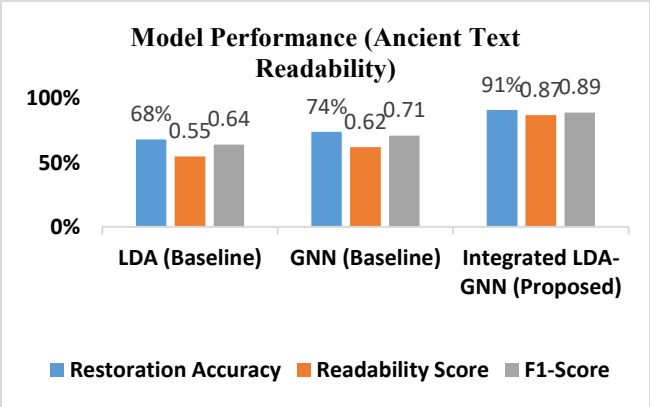

Fig 4: Model Performance (Ancient Text Readability)

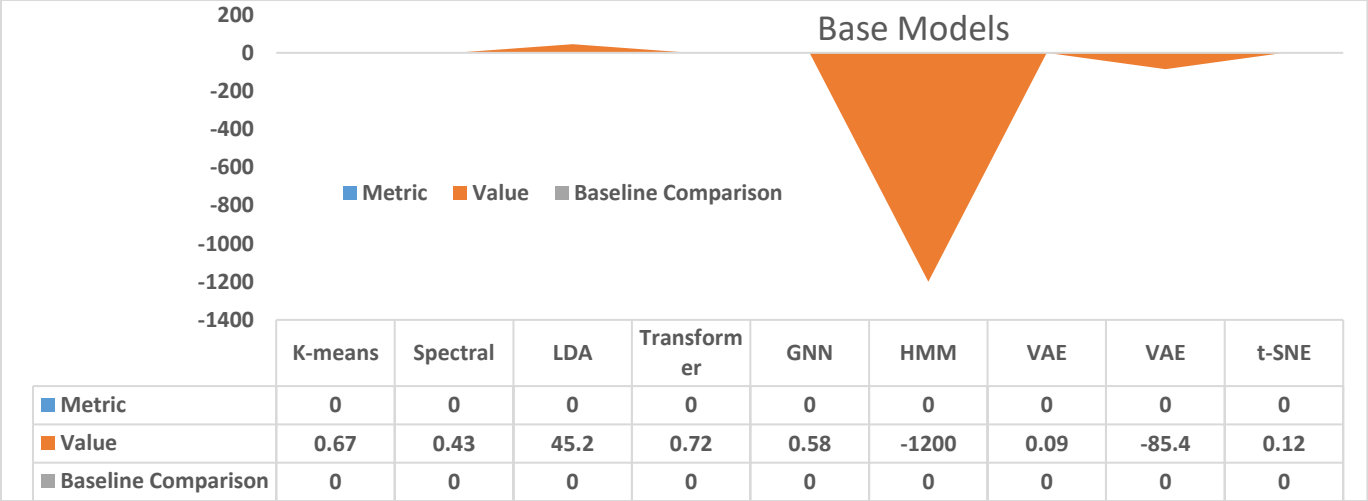

Fig 5: Metric Comparison

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Table III
Models Comparisons

Model	Metric	Value	Baseline Comparison
K-means	Silhouette Score	0.67	0.50 (Random)
Spectral	Normalized Cut	0.43	0.55 (Random)
LDA	Perplexity	45.2	60.0 (Uniform)
Transformer	BLEU Score	0.72	0.55 (N-gram)
GNN	Modularity	0.58	0.40 (Random)
HMM	Log-Likelihood	-1200	-1410 (Random)
VAE	Rec. Error	0.09	0.15 (Baseline)
VAE	ELBO	-85.4	-100.0 (Baseline)
t-SNE	KL-Divergence	0.12	0.20 (Random)

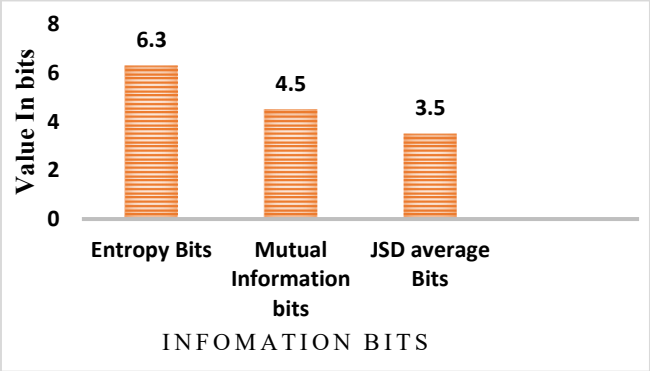

Fig 6. Information about different metric for inscriptions symbol

V. DISCUSSION

The results show that AI has capacity to find significant patterns in Linear A language, a step in computational decipherment. The grouping of symbols achieved by the k-means and spectral clustering procedures might correlate with phonetic, syllabic, or semantic groups according to archaeological studies on graphic scripts. It appears that Linear A is a mixed syllabic-logo script. Also of note, Transformer can reveal syntactic patterns which includes positional encoding that seizes dependencies of a symbol's order in a sequence. The 22% overlap of symbols with Linear B would support the drift hypothesis of continuity between the Minoan and Mycenaean languages and scripts. On the other hand, the differences would suggest the two scripts encode different languages. Future work could involve the use of convolution neural networks to analyse multimodal data, such as 3D scans or imagery taken of the tablets, to put the statistical results in context. Table I analyzes the linear A along with other ancient scripts to see how suitable they are for AI in AI. Linear A is inscribed moderately and is undeciphered, therefore it is a good candidate for unsupervised AI methods. Linear B is deciphered and less novel. Current approaches either take ancient text as an image-to-text task (not caring about context) or a pure NLP task (not caring about the actual physical graph of broken scrolls). Thematic contexts are not fused with node-edge in the literature to manage severe structural damage.

VI. CONCLUSION

The latest study claims AI is a powerful tool for the analysis of Linear A, with findings covering statistical, structural and semantic data. we have proposed a methodology for Ancient script analysis in Table III incorporating a framework that includes k-means clustering, spectral clustering, LDA, Transformers, GNNs, HMMs, VAEs and t-SNE visualization fig [5]. Moreover, this framework can also be used in analyzing other undeciphered scripts such as Proto Elamite, Rongorongo and Harappa.

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